Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/22/2019
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jyagoobi@wpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	508-831-5759
Organization Name:	Worcester Polytechnic Institute		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2016	**End Date:**	09/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
Contact Email:	Mohammad.M.Hasan@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Didion, Jeffrey M.S. (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX16AT09G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism," grant NNX12AR32G, with same Principal Investigator (PI) Jamal Yagoobi, Ph.D. Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism is an International Space Station (ISS) Experiment to investigate the effects of gravity on: 1.) electrically generated two-phase flow and 2.) electrically driven liquid film boiling. The proposed work is a collaborative effort between the investigators to develop and validate the flight experiment hardware. We propose this effort based upon the successes of our current project: NASA Grant/Cooperative Agreement: NNX12AR32G Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism. Although the original effort extends through February 2017, we have completed tasks and achieved the technical and programmatic milestones of		

Task Description:	this project. That is, we have passed all the requirements of the Science Concept Review and received the authorization to advance to the next stage of the experiment development. Therefore, the PI and Co-PI submit this proposal to develop the flight experiment hardware with an effective start date of October 2016. This is a proposal to support the development of the experiment flight hardware for the International Space Station, conduct the experiment on–board the International Space Station, support the operations of the experiment and evaluate the experimental data. The proposed period of performance is October 2016 to September 2021. The robust collaboration between Worcester Polytechnic Institute (WPI) and Goddard Space Flight Center (GSFC) can provide both state of the art theoretical investigation and deliver flight qualified experimental hardware; thus, this proposal presents both investigative and hardware development tasks. The PI will maintain overall technical direction of the experiment development; however, the Co-I will support the flight experimental hardware development in collaboration with Glenn Research Center. Note: This project successfully passed all requirements of the Science Concept Review and has received authorization to advance to the next stage of development. This is to elevate it to the International Space Station flight status as of October 1, 2016, with this continuation grant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are a number of potential terrestrial and space applications for this electrohydrodynamic (EHD) driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices.
Task Progress:	[Ed. note: compiled from PI's FY2018 report for the period September 2017 to July 2018; received June 2019] The following tasks have been accomplished during the period of September 2017 to July 2018. Science Review Document (SRD). The Science Review Document was revised and baselined for the Experiment. Engineering Requirements Document (ERD). The engineering team and investigators developed the Engineering Requirements Document (ERD). Requirements Definition Review (RDR). The investigators and the engineering team conducted a successful Requirements Definition Review on June 7th, 2018 at NASA Glenn Research Center. Preliminary Validation of Experiment Baseplate Design. The investigators successfully conducted a vibration test at Goddard Space Flight Center on March 2nd, 2018 that validated a conceptual experiment disk and heater assembly. This test provided a feasible design for essential experimental hardware that is able to survive launch to International Space Station. Design and Fabrication of Di-Electrophoretic (DEP) Electrode. The investigators designed, fabricated and integrated the DEP electrode into the Science Test Loop at Goddard Space Flight Center. Design, Fabrication and Integration of Science Test Loop. The investigators designed fabricated and integrated the Science Experiment Test Loop at Goddard Space Flight Center to perform a series of experiments to establish the scientific merit of the proposed experiment. The Science Test Loop consisted of the following subsystems: EHD Electrophoretic Pump (EHD Disk), Heater, DEP Electrode, Instrumentation.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Articles in Other Journals or Periodicals	Yang L, Seyed-Yagoobi J. "Liquid-Phase Flow Distribution Control in Meso-Scale with Directionally Reversed Electrohydrodynamic Conduction Pumping Configuration." Journal of Electrostatics. In review as of July 2018. , Jul-2018
Articles in Other Journals or Periodicals	Talmor M, Seyed-Yagoobi J. "Micro-Scale Electrohydrodynamic Conduction Pumping Performance and Heterocharge Layer Morphology Using Forward and Reverse Pumping Orientations." Journal of Electrostatics. In review as of July 2018. , Jul-2018
Books/Book Chapters	Talmor M. Sayed-Yagoobi J. "Electrohydrodynamically Augmented Internal Forced Convection." in "Handbook of Thermal Science and Engineering." Ed. F. Kulacki. Cham: Springer, 2018. p. 429-526. https://doi.org/10.1007/978-3-319-26695-4_7 , Jun-2018
Dissertations and Theses	Yang L. "Flow Distribution Control and Thermal Homogenization with EHD Conduction Pumping and Experimental Studies in Pool Boiling and Internal Condensation." Dissertation, Worcester Polytecnic Institute, Graduation Date: December 2017. , Dec-2017
Papers from Meeting Proceedings	Talmor M, Louste C, Seyed-Yagoobi J. "PIV Flow Field Measurements of Electrohydrodynamic Conduction Pumping." 2018 Electrostatics Joint Conference, Boston, Massachusetts, June 18-20, 2018. Proceedings of the 2018 Electrostatics Joint Conference, Boston, Massachusetts, June 18-20, 2018. http://www.electrostatics.org/images/2018_E7_v2.pdf , Jun-2018
Papers from Meeting Proceedings	Tobar NV, Christidis PN, O'Connor NJ, Talmor M, Seyed-Yagoobi J. "Experimental Study of Flexible EHD Conduction Pumping for Electronics Cooling." Presented at the ASME 2018 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems, San Francisco, California, August 27–30, 2018. Proceedings of the ASME 2018 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems, San Francisco, California, August 27–30, 2018. Paper No. IPACK2018-8322, p. V001T04A012; 6 p. https://doi.org/10.1115/IPACK2018-8322 , Aug-2018

Significant Media Coverage	Yagoobi J (interviewee) "WPI Contributions to Space Travel, Dr. Yagoobi was interviewed about the new EHD cooling technology." Interview by local channel 3, Worcester News Tonight, April 25, 2018. YouTube: https://www.youtube.com/watch?v=8zPUTfVCOIY&t=12s , Apr-2018
Significant Media Coverage	O'Connell S. "WPI prof's cooling technology could aid space travel. Article about Dr. Yagoobi's work with cooling technology." Worcester Telegram, April 23, 2018. https://www.telegram.com/news/20180422/wpi-profs-cooling-technology-could-aid-space-travel , Apr-2018